AI-Driven Cell Culture Automation

Closed-Loop AI System for Automated Cell Culture

UR12e · Elnora AI · Vibrio natriegens

Hackathon March 2026

BACKGROUND

Manual Sub-Culturing Is Broken

Scientists spend hours every day on repetitive, time-critical passaging — and one missed window ruins an experiment.

Time-Dependent

Cells must be sub-cultured at precise growth windows. A few hours late means overgrown or dead cultures.

Human-Dependent

Every passage requires a trained scientist to be physically present — nights, weekends, and holidays included.

Error-Prone & Inconsistent

Manual pipetting, judgment calls on cell density, and fatigue introduce variability across experiments.

THE VISION

A Lab That Runs Itself

24 / 7

Continuous Operation

No shift scheduling needed

0

Human Touchpoints

Fully removed from the loop

100%

Closed-Loop Automation

Sense → Decide → Act

From bench to autonomous bio-factory

UNIVERSAL ROBOTS elnora.

UR12e + Elnora AI can manage the complete cell culture protocol with zero scientists in the loop.

OUR SOLUTION

We Automate the Entire Process

The UR12e robot executes a full passaging protocol autonomously — no human required at any step.

01

Pull Fresh Media

Robot retrieves growth media from fridge

02

Pull Cell Culture

Bacterial cells retrieved from incubator

03

Image on Microscope

Automated brightfield confluency imaging

04

Measure Cell Count

Optical density & cell counter data

05

Decide & Act

Elnora AI triggers next action

Link: https://www.youtube.com/watch?v=xCLaKEvDNW4

DECISION ENGINE

What Triggers a Passage?

INPUT SIGNALS

Microscopic Imaging
Image is taken from microscope and saved on cloud

Cell Count
Image processing through Claude generated image processor that gives cell count and confluence %

Schedule
Elnora AI receives data from Claude to suggest cell culture routine

Contamination Flag
Anomalous morphology detected → terminate culture immediately

elnora.

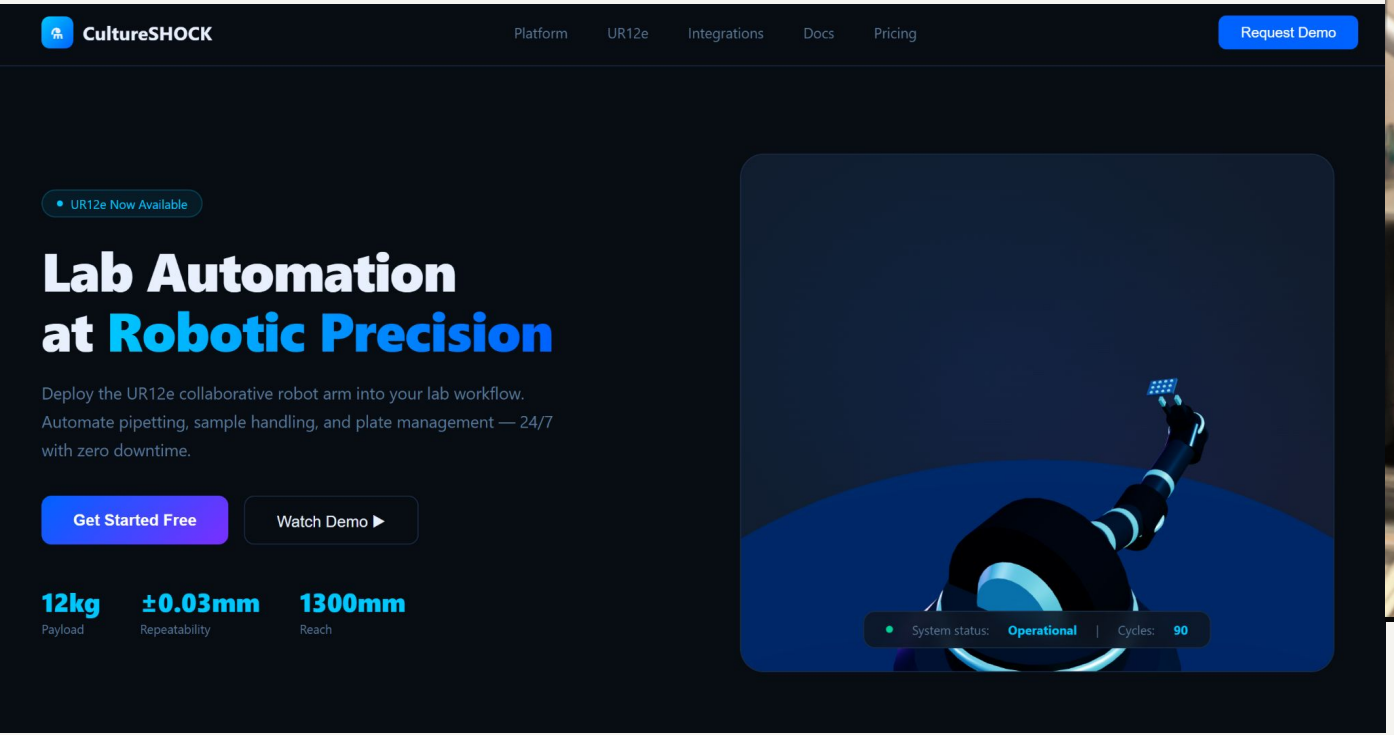
Determines optimal culturing schedule using cell count and confluency data — then commands the robot.

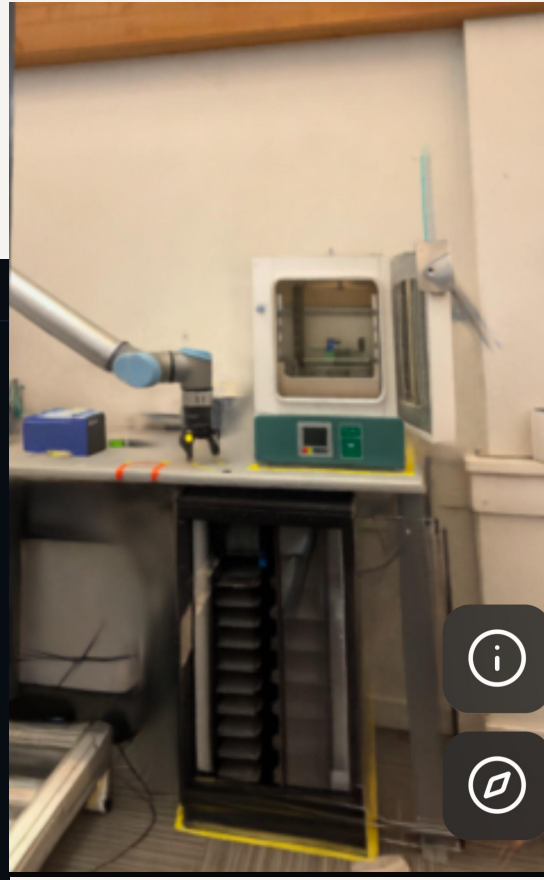
ROBOT ACTIONS

SUBCULTURE
Seed new flask at Elnora-determined density

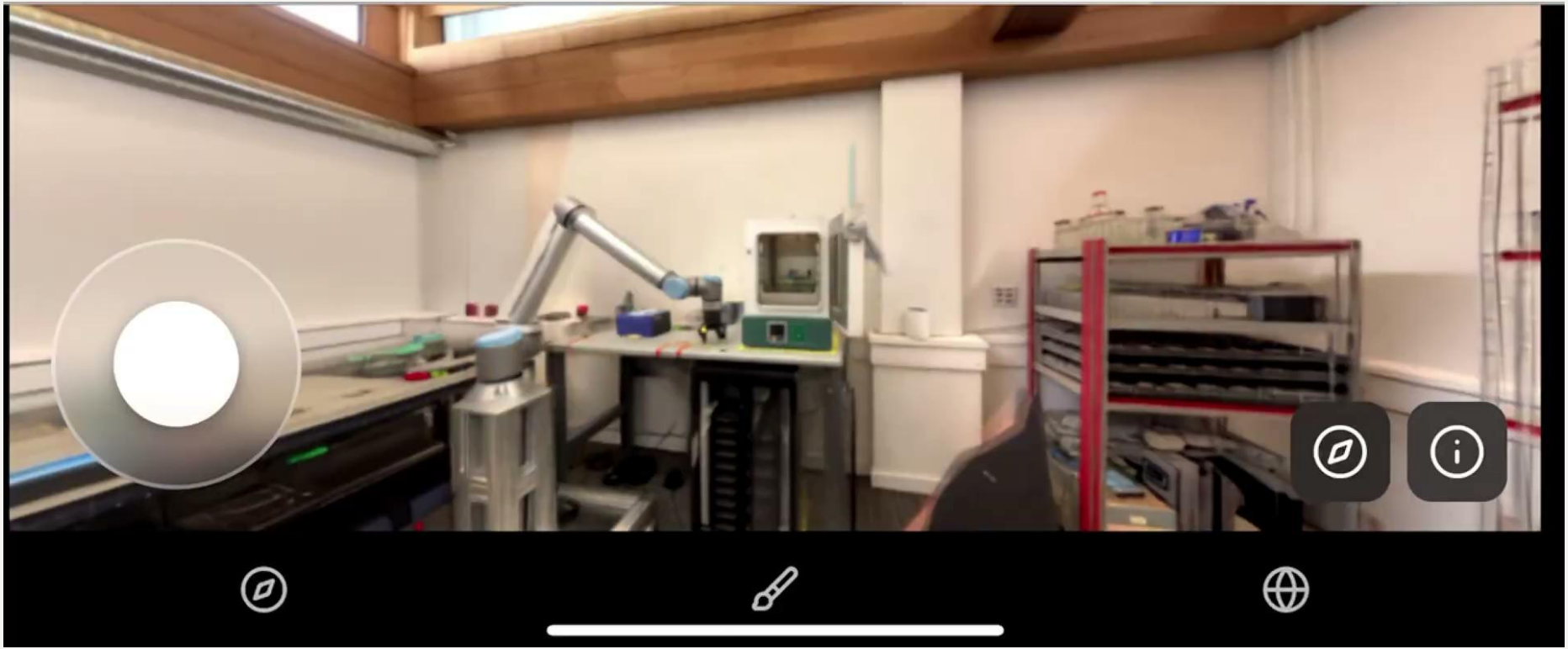
CONTINUE
Metrics within range → incubate & re-check

User Interface for Clients



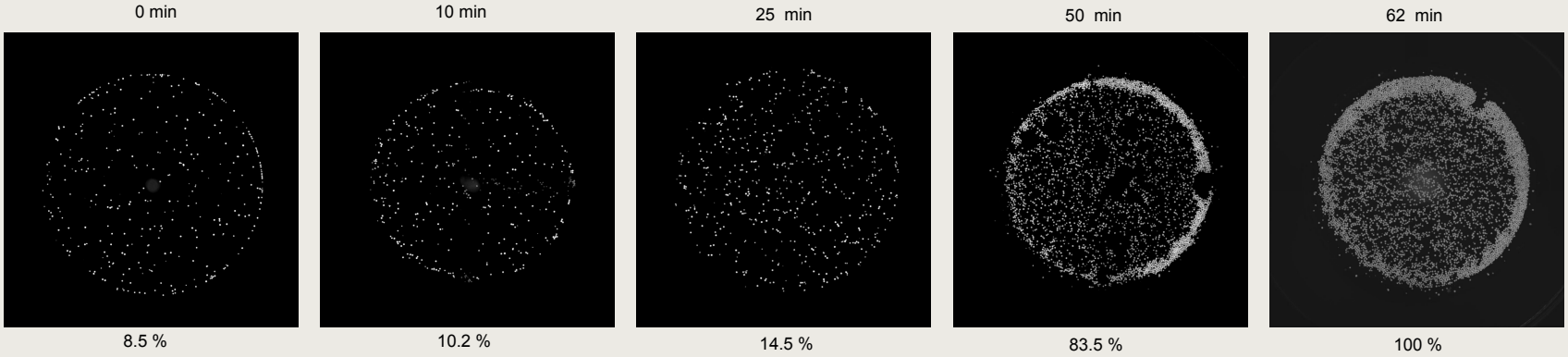


3D Digital Twin



Live 3D Modeling Space

Done in World Labs



Thank You

Recommended subculture time points (to stay in log phase)

- The data show a slow rise to 25 minutes (14.5% confluency), then a sharp jump by 50 minutes (83.5%), reaching 100% at 62 minutes.
- To stay in log phase as much as possible, target **the early part of the steep rise** and avoid the late-log/saturation window.

Suggested time points:

1. **25 minutes** (just before rapid acceleration begins)
2. **30–35 minutes**
3. **40 minutes**
4. **45 minutes**
5. **No later than ~50 minutes**